

Diffuse optics and its application in biomedical imaging

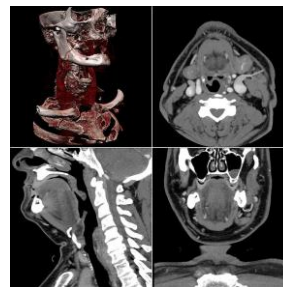
Amir Rosenthal

Imaging – X ray

Radiography



Computed tomography (CT)



Imaging – Ultrasound

2D (analog)



3D (digital)



<http://www.seas.ucla.edu/~scottb/PWave/index.html>

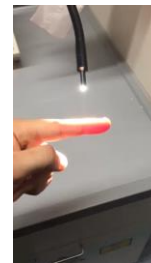
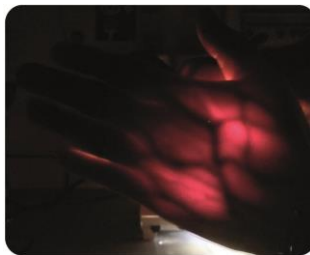
http://blogs.babycenter.com/mom_stories/help-my-unborn-baby-has-a-big-nose

Imaging – Optical

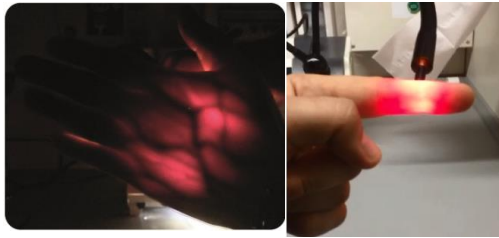
B/W



Color



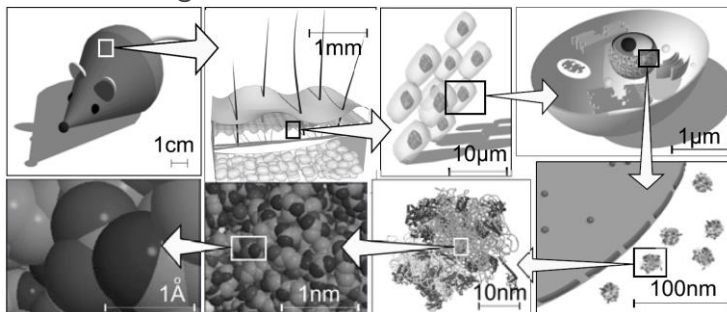
How does light propagate in tissue?



- Diffusion
- “Redshift”
- Local absorption

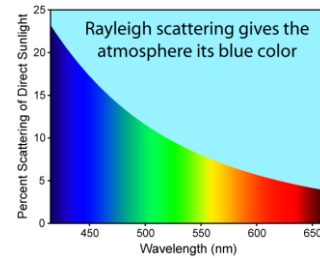
Scattering

Heterogeneous interfaces on all scales

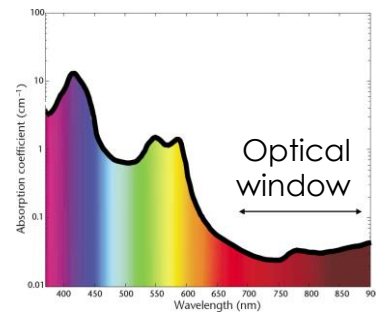
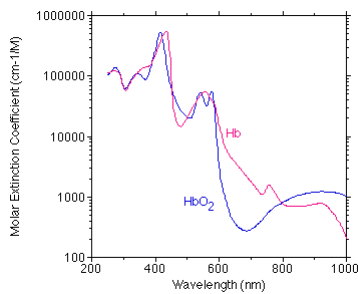


Scattering: general characteristics

- Rayleigh scattering: $I \propto 1/\lambda^4$

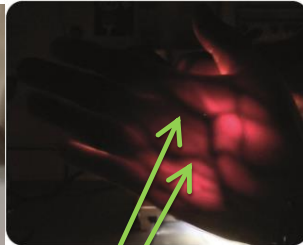


Absorption – mostly caused by blood



<http://omlc.org/spectra/hemoglobin/>
Ntzachristos. Nature Methods 2010

Understanding the optical image



Veins

- ▶ Diffusion due to scattering
- ▶ “Redshift” due to absorption

Course syllabus

- ▶ Microscopic and macroscopic optical properties of biological tissue
- ▶ Light propagation models
- ▶ Planar and tomographic imaging with non-coherent diffuse light
- ▶ Imaging with coherent light: multiple scattering theory and stochastic and deterministic analysis of speckle
- ▶ Hybrid imaging methods

Course philosophy

- ▶ Full mathematical development of the basic equations
- ▶ Explaining basic concepts in the context of real-world applications – what is possible, what not, and why
- ▶ The course is built for a relatively broad audience with background in either medical imaging, image processing, or optics

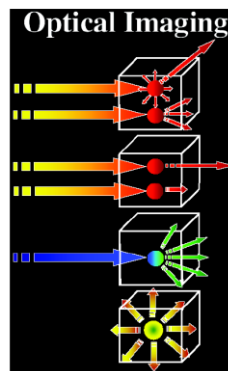
Contrast mechanisms – what is there to see?

Scatter

Absorption
Index of refraction

Fluorescence

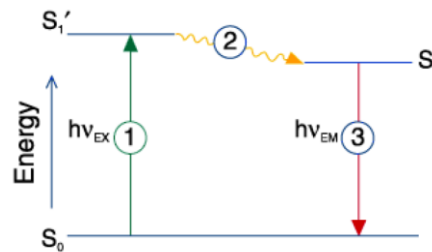
Bioluminescence



Courtesy of Christopher H. Contag, Stanford University

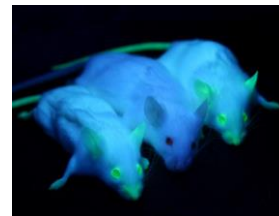
Fluorescence

A quantum process in which a photon is absorbed generates a photon of a different color

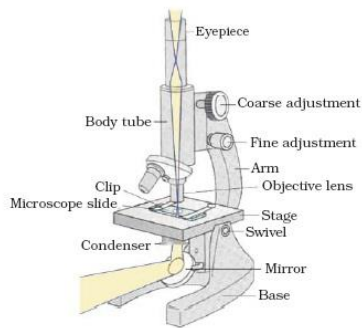


Fluorescent genes – Fluorescent proteins

- ▶ Lab animals may be genetically modified to fluoresce
- ▶ Fluorescence may be done locally (e.g. of specific cells or proteins)
- ▶ 2008 Nobel Prize in Chemistry for the green fluorescent protein (GFP)

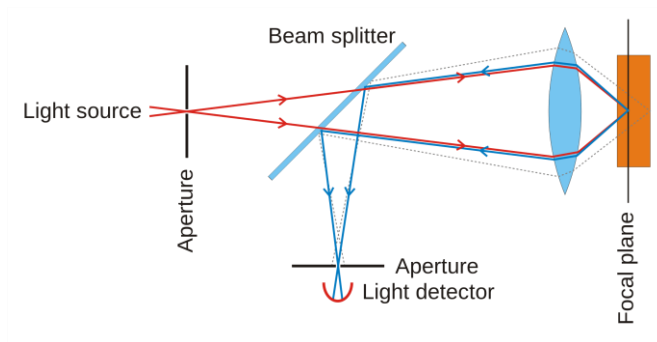


The classical microscope



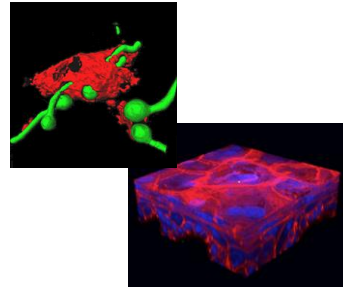
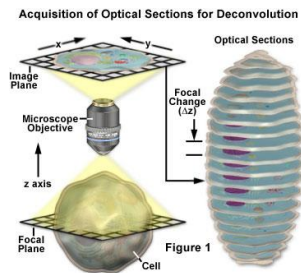
Higher resolution
Deeper images
3D

3D microscopy – confocal



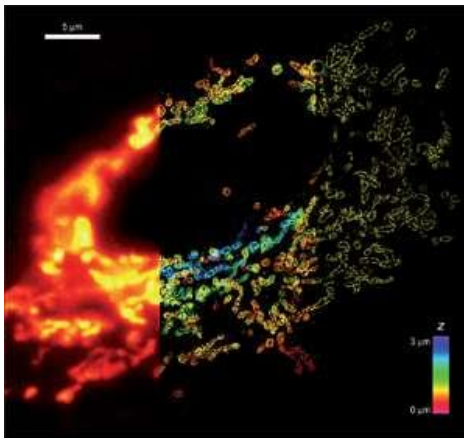
http://en.wikipedia.org/wiki/Confocal_microscopy

3D microscopy – confocal



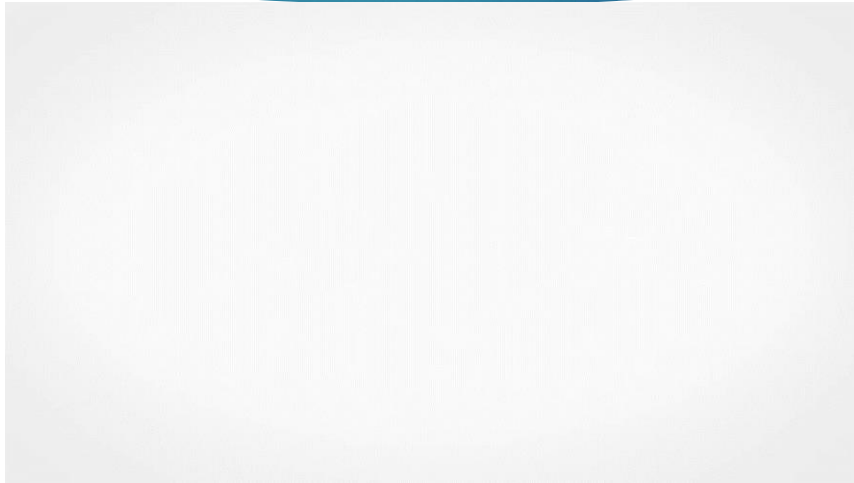
<http://www.olympusmicro.com/primer/digitalimaging/deconvolution/deconintro.html>
 Romani L et al. *Curr Opin Microbiol.* 6:338, 2003
<http://journal.frontiersin.org/article/10.3389/fimmu.2014.00041/full>

Super resolution microscopy



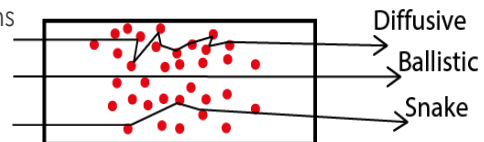
- Imaging smaller than the wavelength
- 2014 Nobel Prize in Chemistry

A multi-disciplinary field

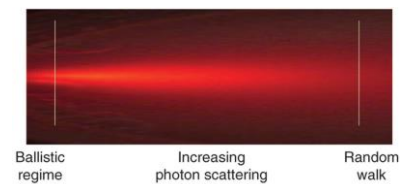


The limits of ballistic imaging

- High-resolution optical imaging requires ballistic/snake photons

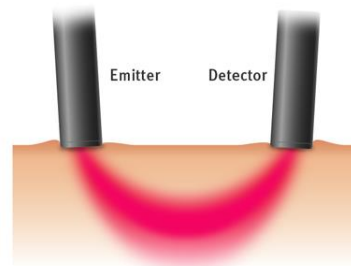


- Deep enough in tissue there are only diffusive photons left

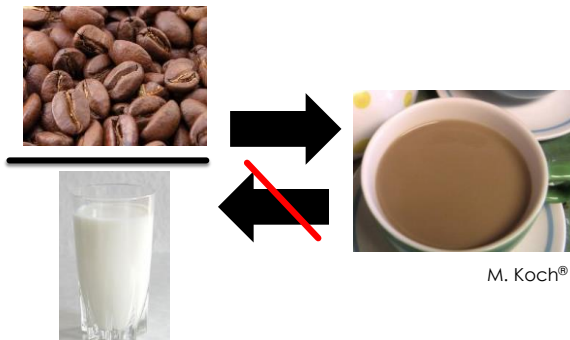


Embracing diffusivity

- The detected light is affected by a large volume of tissue
- Can we measure the average optical properties (absorption, scattering, fluorescence) in that volume?



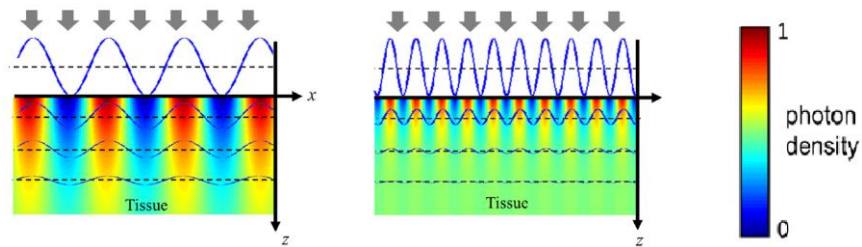
Extracting both scattering and absorption?



M. Koch®

Absorption and scattering cannot be uniquely determined from "simple" photographs

Structured illumination

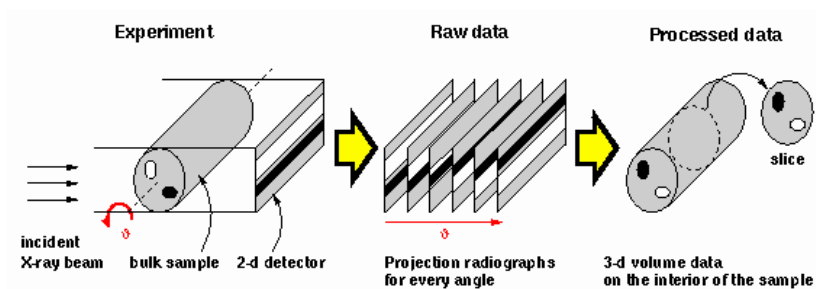


Can measure both scattering and absorption

T. D. O'Sullivan et al. J. Biomed. Opt. 17, 2012.

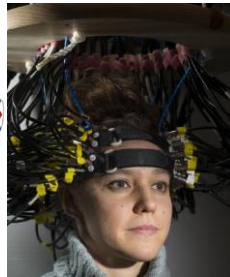
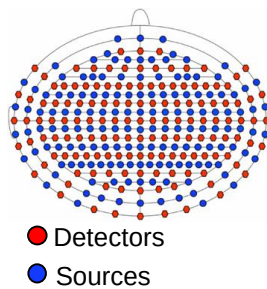
The concept of tomography

Looking at the objects from all sides



<http://www.esrf.eu/UsersAndScience/Experiments/Imaging/IDscisaBomoT/gnigaml/snoitacilppA/22>

Diffuse optical tomography



Diffusion Equation

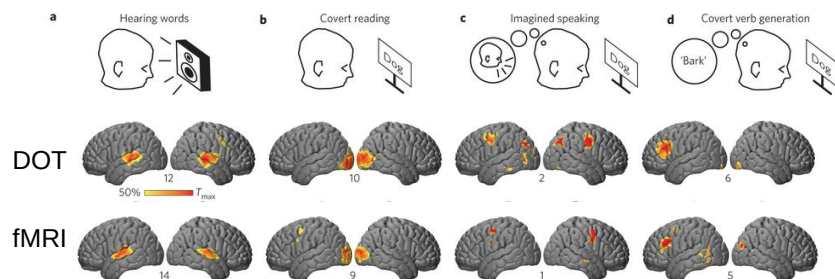
$$-\nabla[D_x(r)\nabla\Phi_x(r)] + \mu_{ax}(r)\Phi_x(r) = S_x(r)$$

$$-\nabla[D_m(r)\nabla\Phi_m(r)] + \mu_{am}(r)\Phi_m(r) = -\Phi_x(r)n(r)$$

The absorption and scattering are found in 3D

K. L. Perdue et al. Phys. Med. Bio. 57, 2012.
A. T. Eggebrecht et al. Nature Photonics 8, 2014.

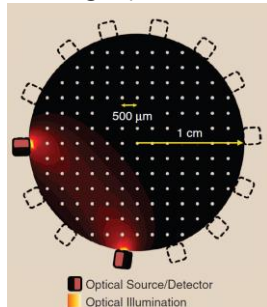
Diffuse optical tomography vs. fMRI



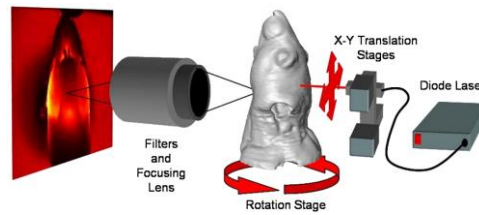
A. T. Eggebrecht et al. Nature Photonics 8, 2014.

Fluorescence molecular imaging

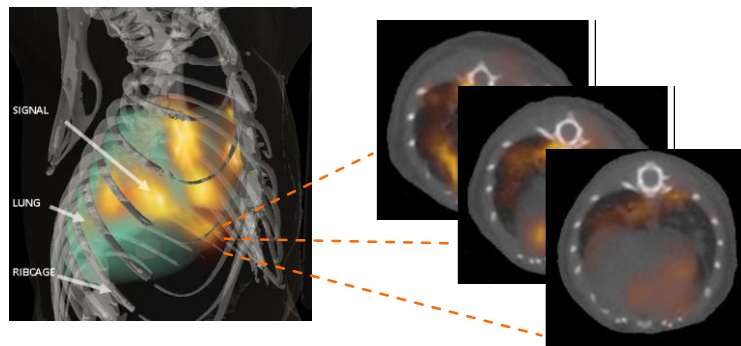
Tomographic view



Camera-based implementations

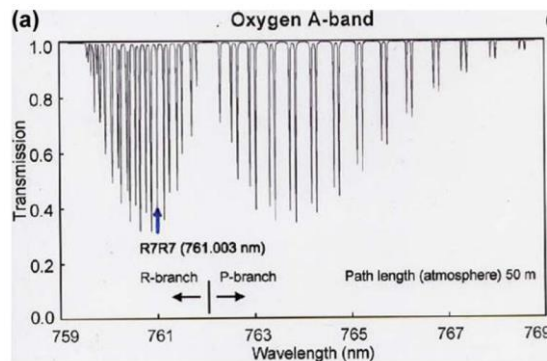


FMT + CT



Absorption spectroscopy

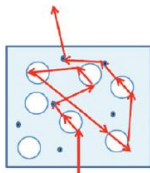
Gases have spectral features much narrower than biological tissue



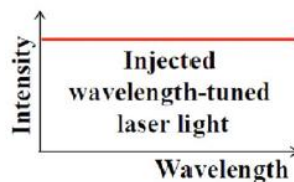
S. Svanberg, Laser Photonics Rev. **7** (2014)

Absorption spectroscopy in a scattering medium

Transmission

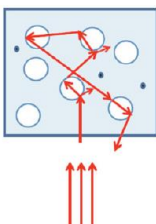


Input spectrum

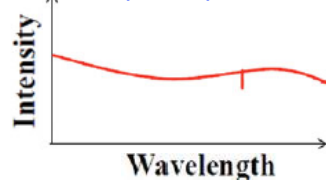


Gas absorption lines are enhanced

Reflection

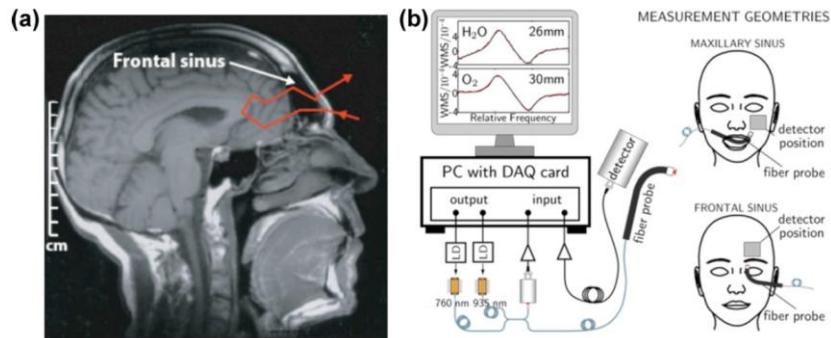


Output spectrum



S. Svanberg, Laser Photonics Rev. **7** (2014)

Spectroscopic probing of frontal sinuses



S. Svanberg, Laser Photonics Rev. **7** (2014)

Imaging of bruises

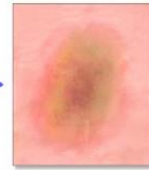
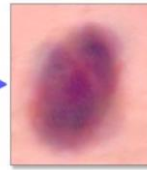


How did it get there?

When did the impact occur?

How do bruises evolve?

Bruises

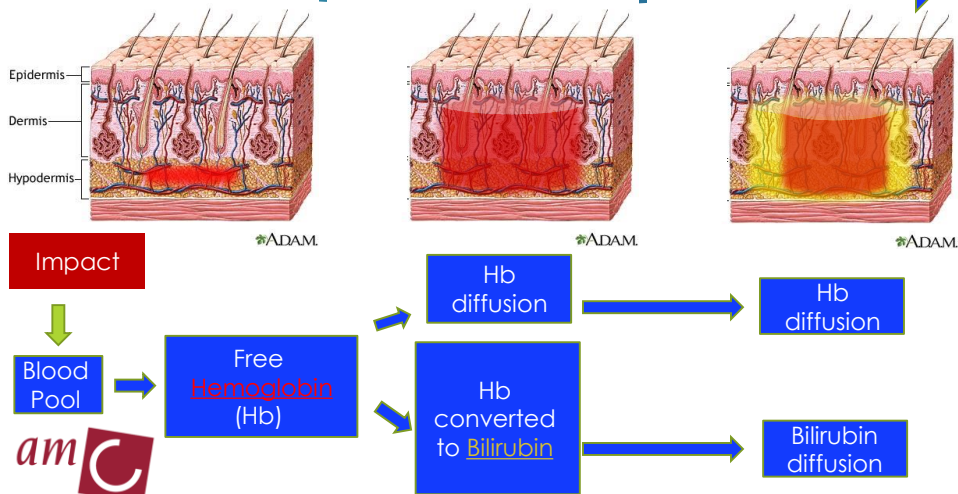


Time

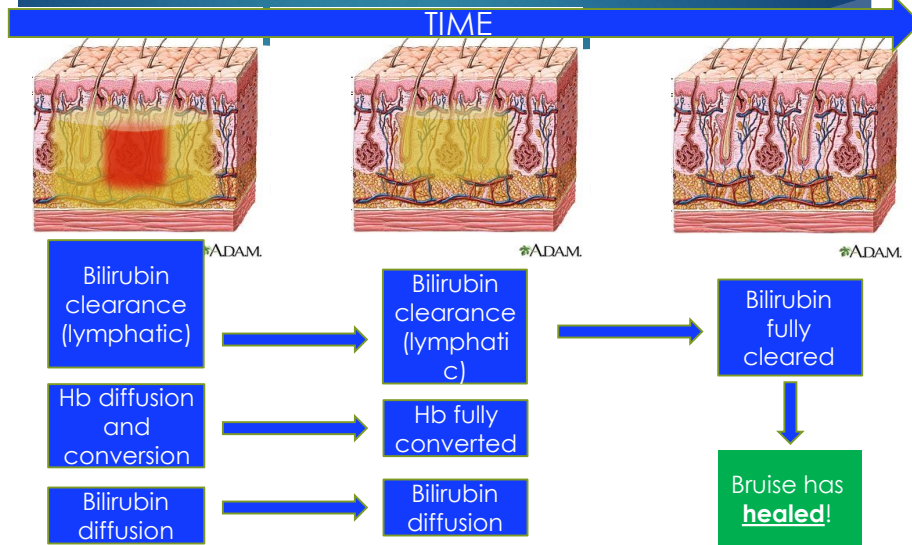


How do bruises evolve?

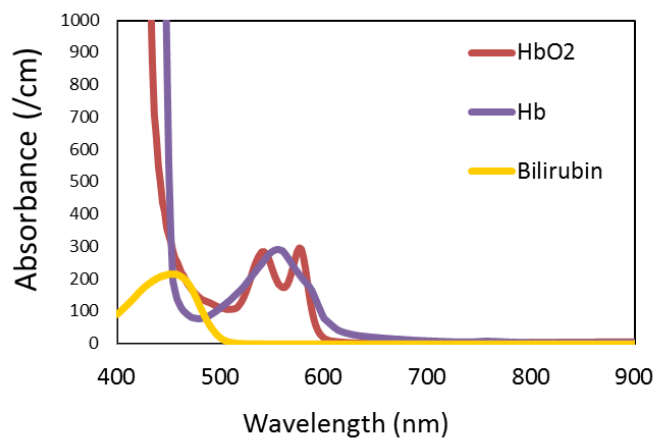
TIME



How do bruises evolve?



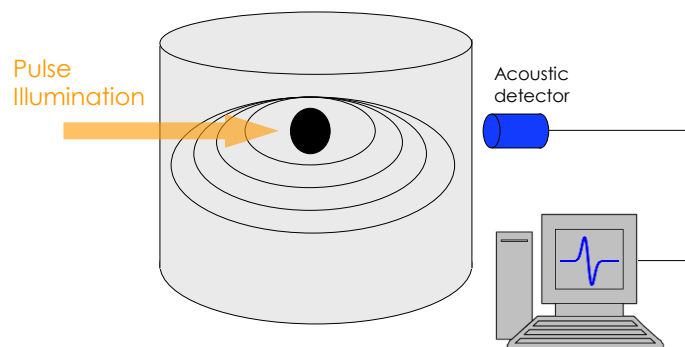
The spectral information



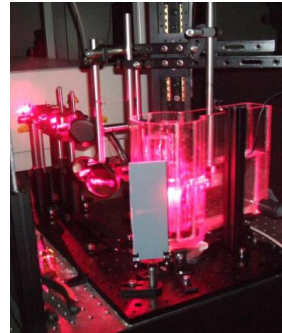
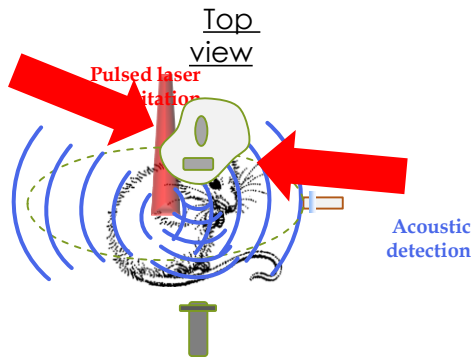
Hybrid imaging

- ▶ Light gets diffused in tissue, ultrasound doesn't
- ▶ Light gives very rich information, ultrasound doesn't
- ▶ Can light and ultrasound be combined to get the best of both worlds?

Optoacoustic signals

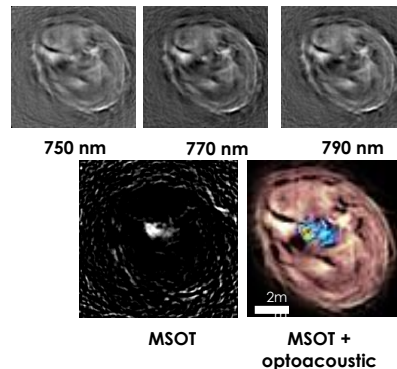
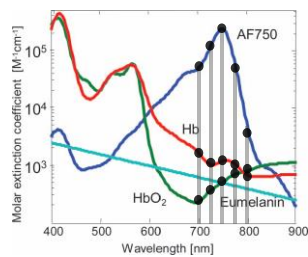


Optoacoustic tomography

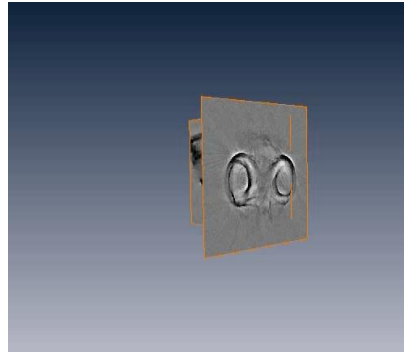
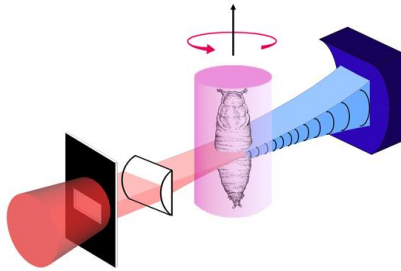


Ma et al., *Optics Exp.*, 17(24), 2009

Multi-spectral optoacoustic tomography (MSOT)

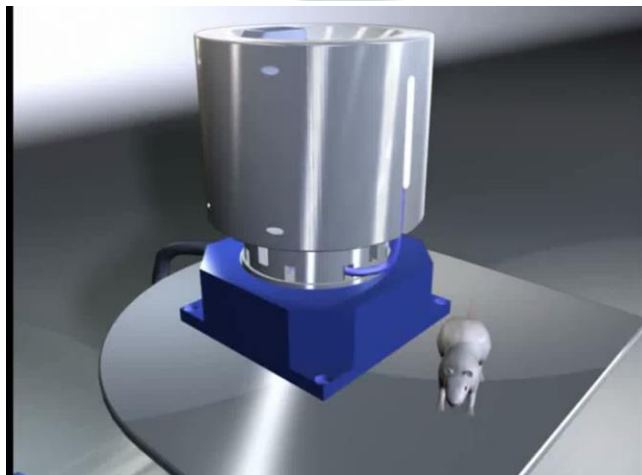


Cross-sectional optoacoustic imaging

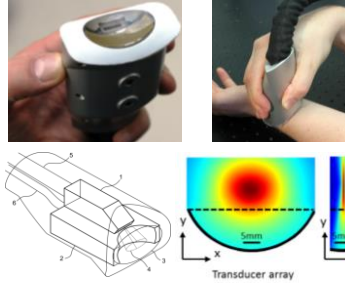


Razansky et al., *Nature Phot.*, 3(7), 2009

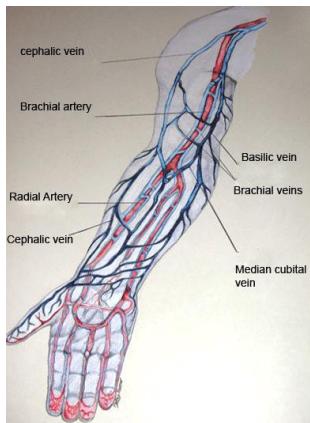
3D optoacoustic imaging



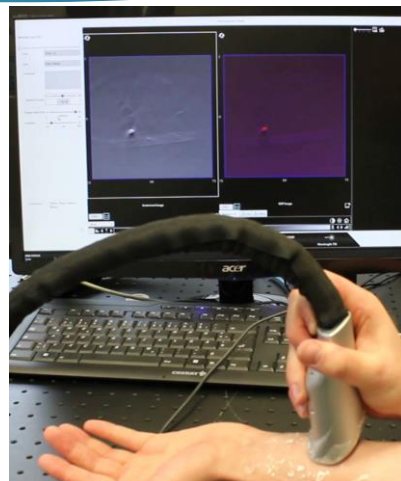
Clinical systems



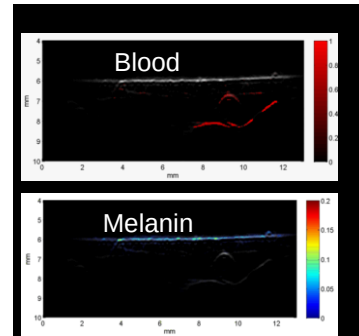
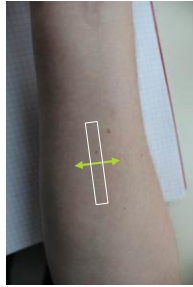
Real-time performance



Buehler *et al.*, Optics Letters, **38**(9), 2013.

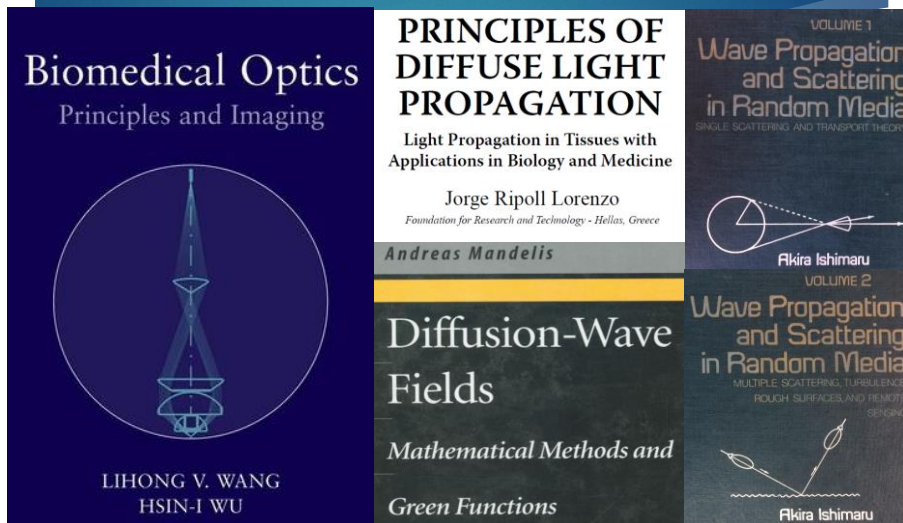


Dynamic distinction between melanin and blood



Buehler *et al.*, Optics Letters, **38**(9), 2013.

Resources



Acknowledgements



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University of Toronto

